Evan Cummings

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Education

Ph.D. Biomedical Engineering, Imaging and Ultrasonics Track **2025**

University of Michigan, Ann Arbor, Michigan

Thesis title: *Rosette Trajectory Cardiac Magnetic Resonance Fingerprinting*

Thesis Advisor: Nicole Seiberlich

B.S. Biomedical Engineering, Devices and Instrumentation Track **2019**

Case Western Reserve University, Cleveland, Ohio

Experience

Seiberlich Lab/Michigan Institute for Imaging Technology and Translation **2017-2026**

Rosette MR Fingerprinting for T_1, T_2, T_2*, and Fat Fraction Mapping

- Developed a trajectory, sequence, and deep image prior reconstruction pipeline for rosette MRF data to simultaneously measure T_1, T_2, T_2*, and fat fraction maps from a single-breathhold sequence to improve non-invasive quantitative assessment of disease
- Optimized acquisition and reconstruction techniques for cardiac, liver, and carotid applications across 0.55T, 1.5T, and 3T systems
- Techniques reduced required breathhold duration, improved spatial resolution, and reduced quantitative errors compared to previous approaches.
- Presented findings at ISMRM, RSNA, and SCMR, and published in Magnetic Resonance in Medicine

Multipoint Reconstruction for Radial GRAPPA

- Worked with the through-time radial GRAPPA reconstruction algorithm to improve the speed and memory requirements for reconstructing highly undersampled data, giving clinicians more freedom to switch slice orientations during a real-time scan or interventional procedures
- Approach reduced calibration time from 30s to 5s with minimal impact on image quality
- Presented findings at conferences, and published in Magnetic Resonance in Medicine

Siemens Healthineers Collaboration **2023**

MR Fingerprinting Development Kit

- Worked with the Siemens development team onsite in Erlangen to implement and test 3D prostate and free-running cardiac MRF sequences in the MRFDK framework
- Developed documentation and presentations on the use of the toolkit, and made a recommendation to Seiberlich lab whether to integrate it into the lab's development workflow

Technical Skills

- Mathematics: Proficient working with inverse problems and optimization algorithms
- Pulse Sequence Development: Proficient with C++ and Siemens IDEA environment
- Image Reconstruction: Proficient working with parallel imaging, compressed sensing, and machine learning-based reconstruction algorithms in MATLAB, Python, and Tensorflow

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Additional Skills

- Collaborations: Worked with colleagues internally to help adapt my techniques for application in the liver and other organs, worked with external partners on collaborative development of software tools, and worked with clinician scientists to deploy and evaluate techniques in patient studies.
- Communication: Experienced in written communication in the form of prepared abstracts and manuscripts, and have presented research at 6 international conferences.

Publications

Cummings E, Cruz G, Kaplan S, Hamilton J, Seiberlich N. IDEAL-DIP reconstruction enables rosette trajectory MR fingerprinting at 0.55T for single-breathhold T_1, T_2, T_2*, and fat fraction mapping. [manuscript in preparation]

Cummings E, Cruz G, Richardson J, Kaplan S, Hamilton J, Seiberlich N. Rosette cardiac MR fingerprinting for simultaneous T_1, T_2, T_2*, and fat fraction mapping using a multi-echo deep image prior reconstruction. *Magn Reson Med.* 2026; doi:10.1002/mrm.70299.

Ahad J, Cummings E, Franson D, Hamilton J, Seiberlich N. Optimization of through-time radial GRAPPA with coil compression and weight sharing. *Magn Reson Med.* 2022;88:1244-1254

Cummings E, MacDonald JA, Seiberlich N. Parallel Imaging. In: Prieto C, Doneya M, Akacaya C, *MRI Reconstruction: Theory, Methods, and Applications.* Elsevier; 2021:129-157

Selected Conference Presentations

Cummings E, Cruz G, Hamilton J, Seiberlich N. Single-breathhold Cardiac T_1, T_2, T_2*, and fat fraction mapping at 0.55T using dual rosette trajectory MRF and a deep image prior reconstruction. *International Society for Magnetic Resonance in Medicine Annual Meeting,* Honolulu, 2025.

Cummings E, Cruz G, Hamilton J, Seiberlich N. Cardiac T_1, T_2, T_2*, and PDFF Mapping at 0.55T with Rosette MR Fingerprinting and a Deep Image Prior Reconstruction. *Society of Cardiac Magnetic Resonance Annual Meeting,* Washington DC, 2025.

Cummings E, Cruz G, Kaplan S, Richardson J, Hamilton J, Seiberlich N. Cluster-based Low-rank Regularization for Cardiac Rosette MR Fingerprinting. *International Society for Magnetic Resonance in Medicine Annual Meeting,* Singapore, 2024.